① PPL original

$$\min z = 4x_1 + x_2$$

s.a.

$$\begin{aligned} 3x_1 + x_2 &= 3 \\ 4x_1 + 3x_2 &\geq 6 \\ x_1 + 2x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

② PPL aumentado

$$\min z = 4x_1 + x_2$$

s.a.

$$\begin{aligned} 3x_1 + x_2 &= 3 \\ 4x_1 + 3x_2 - x_3 &= 6 \\ x_1 + 2x_2 + x_4 &= 4 \\ x_i &\geq 0, \quad i = 1, \ldots, 4. \end{aligned}$$

③ PPL artificial

$$\min z = 4x_1 + x_2 + M\bar{x}_5 + M\bar{x}_6 \iff \max -z = -4x_1 - x_2 - M\bar{x}_5 - M\bar{x}_6$$

$$\Updownarrow$$

$$\max -z + 4x_1 + x_2 + M\bar{x}_5 + M\bar{x}_6 = 0$$

s.a.

$$\begin{aligned} 3x_1 + x_2 + \bar{x}_5 &= 3 \\ 4x_1 + 3x_2 - x_3 + \bar{x}_6 &= 6 \\ x_1 + 2x_2 + x_4 &= 4. \end{aligned}$$

$$x_1, x_2, x_3, x_4, \bar{x}_5, \bar{x}_6 \geq 0$$

V.B	x_1	x_2	x_3	x_4	$\bar{x}_5$	$\bar{x}_6$	RHS	
$-z$	4	1	0	0	M	M	0	
$-z$	$4-7M$ ↓	$1-4M$	M	0	0	0	$-9M$	$L_0 = L_0 - M(L_1 + L_2)$
← $\bar{x}_5$	3	1	0	0	1	0	3	$\min\left\{\frac{3}{3}, \frac{6}{4}, \frac{4}{1}\right\} = 1$
$\bar{x}_6$	4	3	−1	0	0	1	6	
x_4	1	2	0	1	0	0	4	
$-z$	0	$\frac{-1-5M}{3}$ ↓	M	0	$\frac{7M-4}{3}$	0	$-2M-4$	$L_0^{(1)} = L_0^{(0)} + (7M-4)L_1^{(1)}$
x_1	1	1/3	0	0	1/3	0	1	$L_1^{(1)} = L_1^{(0)}/3$
← $\bar{x}_6$	0	(5/3)	−1	0	−4/3	1	2	$L_2^{(1)} = L_2^{(0)} - 4L_1^{(1)}$
x_4	0	5/3	0	1	−1/3	0	3	$L_3^{(1)} = L_3^{(0)} - L_1^{(1)}$

$$\min\left\{\frac{1}{\frac{1}{3}}, \frac{2}{\frac{5}{3}}, \frac{3}{\frac{5}{3}}\right\} = \frac{6}{5}$$

V.B	x_1	x_2	x_3	x_4	$\bar{x}_5$	$\bar{x}_6$	RHS	
$-z$	0	0	−1/5 ↓	0	$\frac{5M-8}{5}$	$\frac{1+5M}{5}$	$-18/5$	$L_0^{(2)} = L_0^{(1)} + \left(\frac{1+5M}{3}\right)L_2^{(2)}$
x_1	1	0	1/5	0	3/5	−1/5	3/5	$L_1^{(2)} = L_1^{(1)} - \frac{1}{3}L_2^{(2)}$
x_2	0	1	−3/5	0	−4/5	3/5	6/5	$L_2^{(2)} = \frac{3}{5}L_2^{(1)}$
← x_4	0	0	(1)	1	1	−1	1	$L_3^{(2)} = L_3^{(1)} - \frac{5}{3}L_2^{(2)}$
$-z$	0	0	0	1/5	$\frac{5M-7}{5}$	M	$-17/5$	$L_0^{(3)} = L_0^{(2)} + \frac{1}{5}L_3^{(3)}$
x_1			0				3/5	$L_1^{(3)} = L_1^{(2)} - \frac{1}{5}L_3^{(3)}$
x_2			0				9/5	$L_2^{(3)} = L_2^{(2)} + \frac{3}{5}L_3^{(3)}$
x_3	0	0	1	1	1	−1	1	$L_3^{(3)} = L_3^{(2)}$

$$x^* = \begin{bmatrix} 3/5 \\ 9/5 \\ 1 \\ 0 \end{bmatrix} \quad \text{e} \quad z^* = 17/5$$